

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch :Civil Semester :V Subject Code & Name: BTCVPE506G Material, Testing and Evaluation Max Marks: 60 Date:16/02/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Define ferrous metals and non-ferrous metals with examples? Explain about ferrous metals	CO1	6
B)	Write a note on Geo-synthetics.	CO2	6
C)	With the help of stress-strain diagram, explain strain hardening	CO2	6
Q.2	Solve Any Two of the following.		12
A)	Differentiate between fiber reinforced concrete and steel reinforced concrete?	CO2	6
B)	Enumerate the characteristics of good timber	CO3	6
C)	Explain any one non-destructive method of testing of concrete	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Why proportioning of concrete is necessary.	CO1	6
B)	How does hardening of cement occur? Explain.	CO1	6
C)	Explain the advantage of sand –lime brick.	CO1	6
Q.4	Solve Any Two of the following.		12
A)	Write in detail the tensile test procedure of mild steel.	CO2	6
B)	Elaborate the term -3D printing.	CO2	6
C)	Give the classification of glass.	CO1	6
Q. 5	Solve Any Two of the following.		12
A)	What do you mean by composite material? Explain about the composition of composite materials.	CO1	6

B)	What are properties of FRP?	CO1	6
C)	Give the effect of improper curing of concrete.	CO2	6
	*** End ***		